Nischal K Shaj

Full Stack Developer

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Leetcode

Profile Summary

Backend-focused Full-Stack Developer experienced in building production applications with Node.js, Express.js, TypeScript, and PostgreSQL. Skilled in REST API development, database optimization, authentication, real-time systems, and cloud deployment, with experience across enterprise and independent business projects.

Experience

- **Web Developer (Backend Engineering) at Scita Solutions** *Dec 2024 – Present*
  - Built and optimized Node.js/Express APIs with PostgreSQL, improving request performance by approximately 30%.
  - Implemented JWT authentication and centralized validation/error-handling middleware for production APIs.
  - Optimized SQL joins and indexes for reporting dashboards; supported client deployments and production issue resolution.
  - Collaborated with product and frontend teams on feature delivery and API integration.

Technical Skills

- Languages: JavaScript, TypeScript
- Frontend: React, Next.js, Tailwind CSS
- Backend: Node.js, Express.js, REST APIs, JWT Authentication, Socket.IO
- Databases: MongoDB, PostgreSQL
- Integrations: Cloudinary, Nodemailer, Razorpay, PayPal, Firebase, Multer
- Infrastructure and Deployment: Docker, Nginx, PM2, CI/CD, AWS EC2, Vercel, Render, GitHub Actions
- Core: Data Structures and Algorithms, System Design

Projects

Scita ATMS – Advanced Traffic Management System

- Developed an Advanced Traffic Management System aligned with **NHAI ATMS 2023 guidelines** for highway surveillance and enforcement.
- Integrated **ANPR, VIDS, overspeed detection, incident logging, and VMS control**, with real-time alerts and device monitoring.
- Built module-wise reporting with downloadable reports and evidence archives using **PDFKit**.
- **Tech: Node.js, Express.js, PostgreSQL, Socket.IO, PM2**

Mangaluru Frosty – Business Product Catalogue | Pro Bono Live

- Developed a responsive product catalogue for customers to browse available products and pricing.
- Built an admin dashboard with product CRUD operations and environment-based login authentication.
- Integrated **Cloudinary** for product image storage and **WhatsApp** for customer-initiated orders using pre-filled messages.
- Implemented a layered Express.js backend with MongoDB and Nodemailer email communication.
- **Tech: Next.js, TypeScript, Node.js, Express.js, MongoDB, Cloudinary, Nodemailer**

Code Sprint – E-Learning Platform Frontend GitHub | Backend GitHub

- Developed a programming and competitive coding platform with live courses and interactive learning modules.
- Integrated payment gateways, subscription tiers, remote code execution, and real-time chat using Socket.IO.
- Built admin analytics and implemented CI/CD and containerized AWS deployment using **Docker, NGINX, and GitHub Actions**.
- **Tech: Node.js, Express.js, MongoDB, React, Next.js, AWS, S3, Socket.IO**

Education

MERN Stack Development 2023 - 2024
Brototype, Trivandrum

B.Sc Cyber Forensic 2020 - 2023
School of Technology and Applied Science, Pathanamthitta, Kerala
CGPA: 7.61